

Validating AI for Image Analysis

Part 1: Metrics

Dr. Annika Reinke, Dr. Evangelia Christodoulou

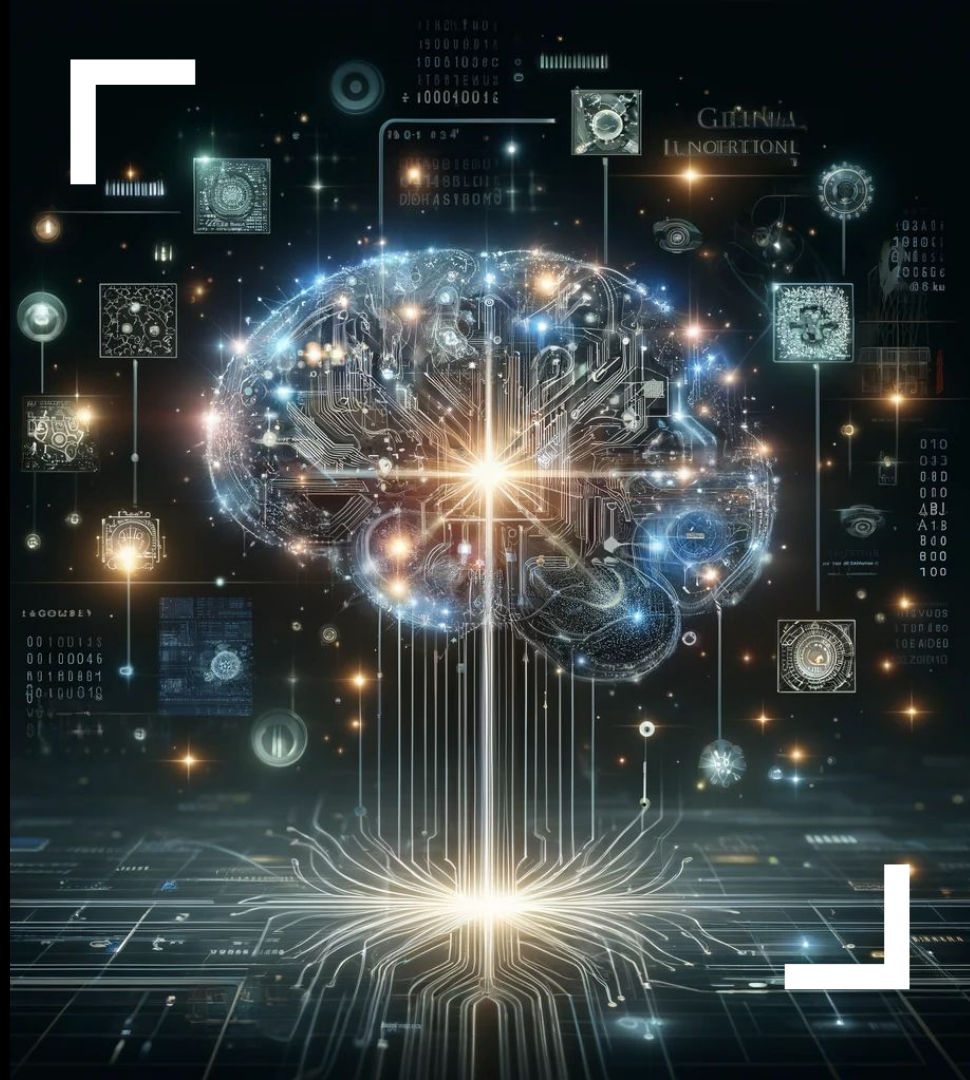
Div. Intelligent Medical Systems division, German Cancer Research Center (DKFZ)

Helmholtz Imaging

 HELMHOLTZ
IMAGING


Research for a Life without Cancer

GERMAN
CANCER RESEARCH CENTER
IN THE HELMHOLTZ ASSOCIATION



Who we are



Dr. Annika Reinke

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Co-Group Lead "Validation of Intelligent
Systems"

'Meet & Greet'



Icebreaker:

Which metrics do you frequently
use?



Agenda & General Remarks (June 3th 2026)



Time	Topic
09:00	Introduction
09:15	Classification essentials and practicals
10:00	Segmentation essentials and practicals
10:30	BREAK
10:50	Metrics Reloaded and practicals
12:00	End of lecture 1

Remark 1: If not done so already, please pull from our Github repository

Remark 2: Please keep menti.com open. We will use it throughout the lecture



Scope of today



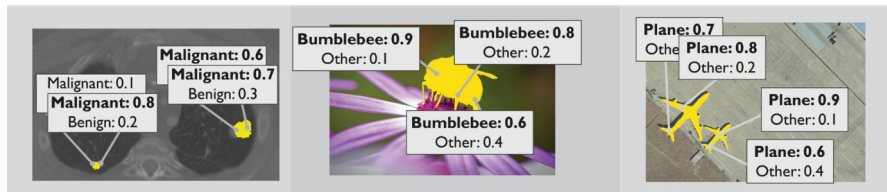
Image-level classification

Assign class label(s) per image



Segmentation

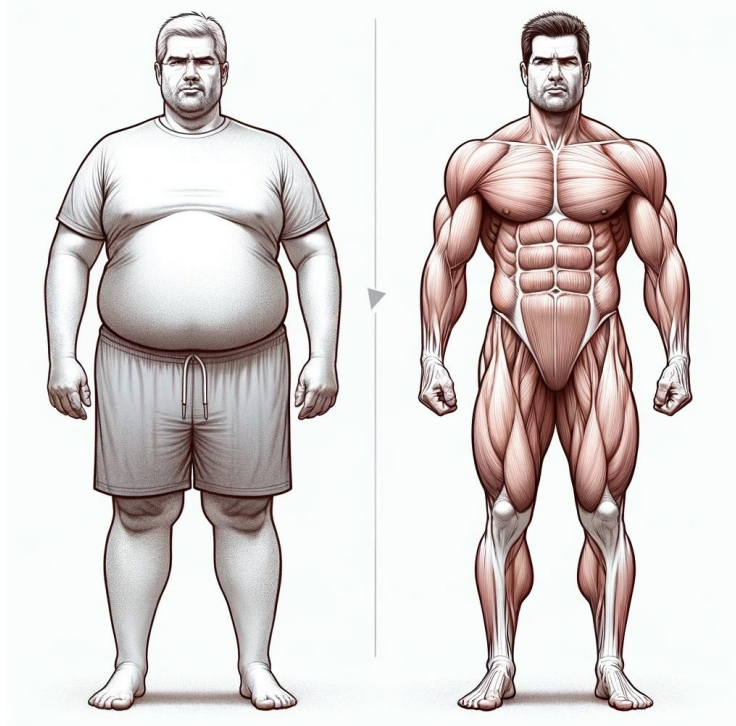
Assign class label(s) per pixel/voxel



"Metrics": key to trustworthy performance assessment



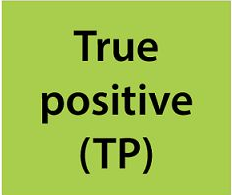
100 kg

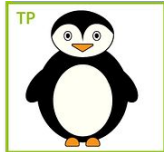
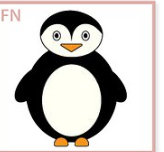
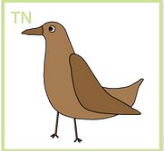
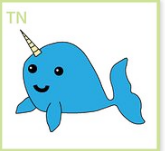
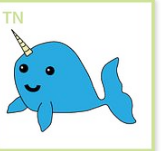
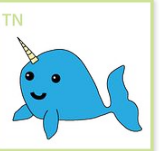
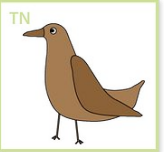
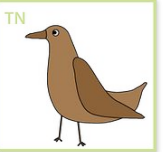
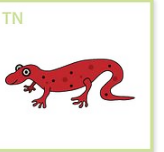
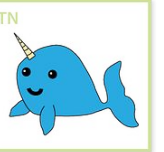
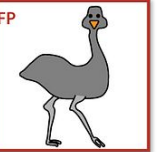


100 kg

Essentials for classification

The confusion matrix

		PREDICTED	
		Positive	Negative
ACTUAL	Positive	 True positive (TP)	 False negative (FN)
	Negative	 False positive (FP)	 True negative (TN)

 TP Prediction: Penguin ✓	 TP Prediction: Penguin ✓	 TP Prediction: Penguin ✓	 TP Prediction: Penguin ✓	 TP Prediction: Penguin ✓
 TP Prediction: Penguin ✓	 TP Prediction: Penguin ✓	 FN Prediction: No penguin ✗	 FN Prediction: No penguin ✗	 TN Prediction: No penguin ✓
 TN Prediction: No penguin ✓	 TN Prediction: No penguin ✓	 TN Prediction: No penguin ✓	 TN Prediction: No penguin ✓	 TN Prediction: No penguin ✓
 TN Prediction: No penguin ✓	 TN Prediction: No penguin ✓	 TN Prediction: No penguin ✓	 TN Prediction: No penguin ✓	 FP Prediction: Penguin ✗



The confusion matrix for multiple classes

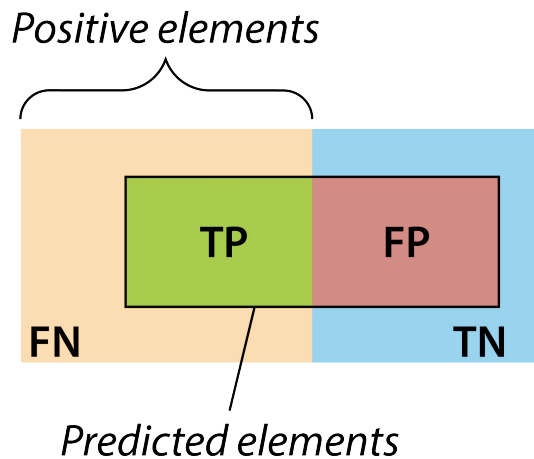


		Predicted				
Actual		1	2	...	C	Σ
	1	n_{11}	n_{12}	...	n_{1C}	$n_{1.}$
	2	n_{21}	n_{22}	...	n_{2C}	$n_{2.}$
	...	...	...	...	...	...
	C	n_{C1}	n_{C2}	...	n_{CC}	$n_{C.}$
	Σ	$n_{.1}$	$n_{.2}$	...	$n_{.C}$	N

Accuracy



$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

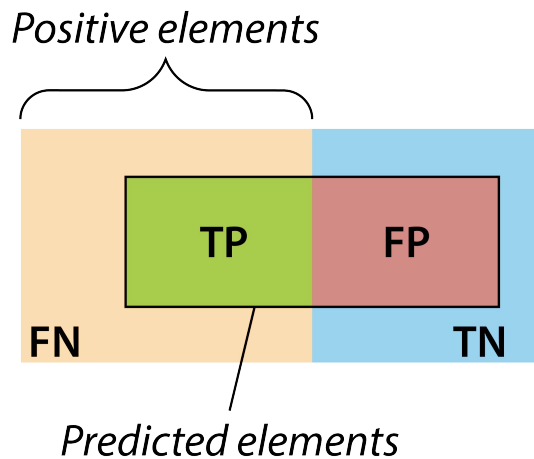


$$\text{Accuracy} = \frac{\text{[Diagram: TP and TN boxes]}}{\text{[Diagram: All four boxes (TP, FP, TN, FN)]}}$$

Balanced Accuracy (BA)



$$BA = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) = \frac{1}{2} (\text{Sensitivity} + \text{Specificity})$$



$$BA = \frac{1}{2} \left(\frac{\text{TP}}{TP + FN} + \frac{TN}{TN + FP} \right)$$

Practical exercise 1.1

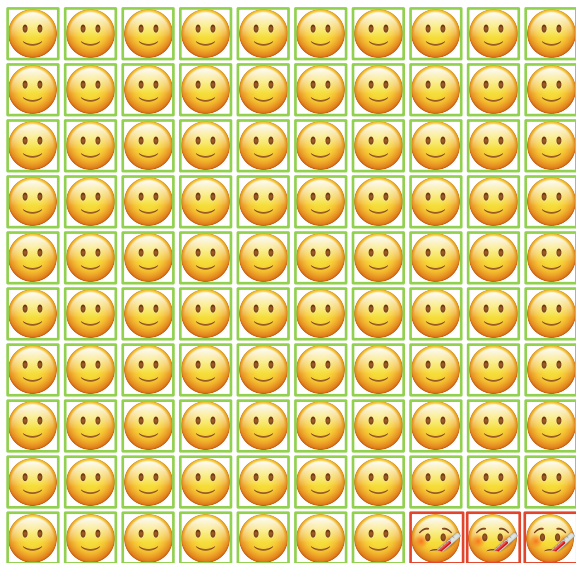


Practical 1.1

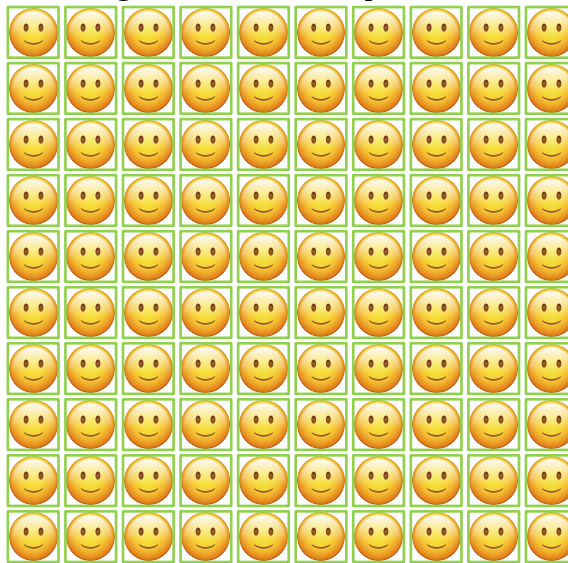


Goal: Classify whether patients into 'cancer' or 'no cancer' (n=100)

Dataset



Algorithm output



$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

$$\text{BA} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) = \frac{1}{2} (\text{Sensitivity} + \text{Specificity})$$

Practical exercise 1.2



Practical 1.2

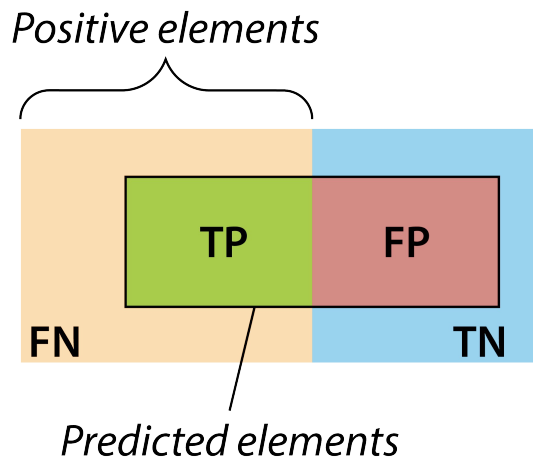


		PREDICTED	
		Positive	Negative
ACTUAL	Positive	TP 15	FN 5
	Negative	FP 4,980	TN 95,000

Positive Predictive Value (PPV) / Precision

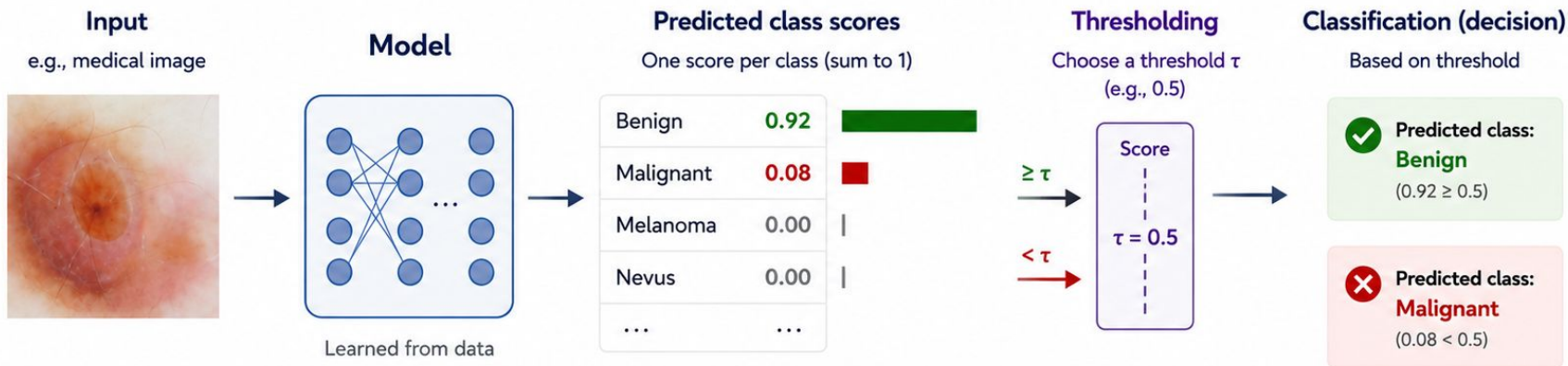


$$\text{PPV} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$



$$\text{PPV} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Predicted class scores



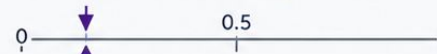
Effect of the threshold τ (binary example: Malignant vs. Not malignant)

High threshold (e.g., $\tau = 0.8$)



- + Fewer false positives (high specificity)
- More false negatives (lower sensitivity)

Low threshold (e.g., $\tau = 0.2$)



- + More true positives (high sensitivity)
- More false positives (lower specificity)

Essentials for segmentation

Classification vs. segmentation



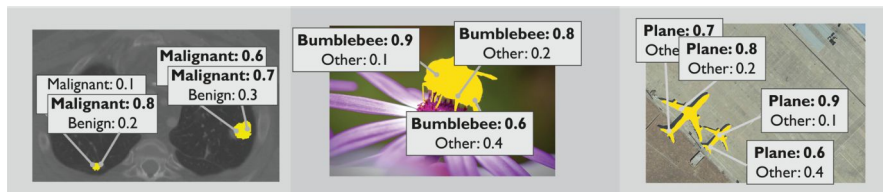
Image-level classification

Assign class label(s) per image

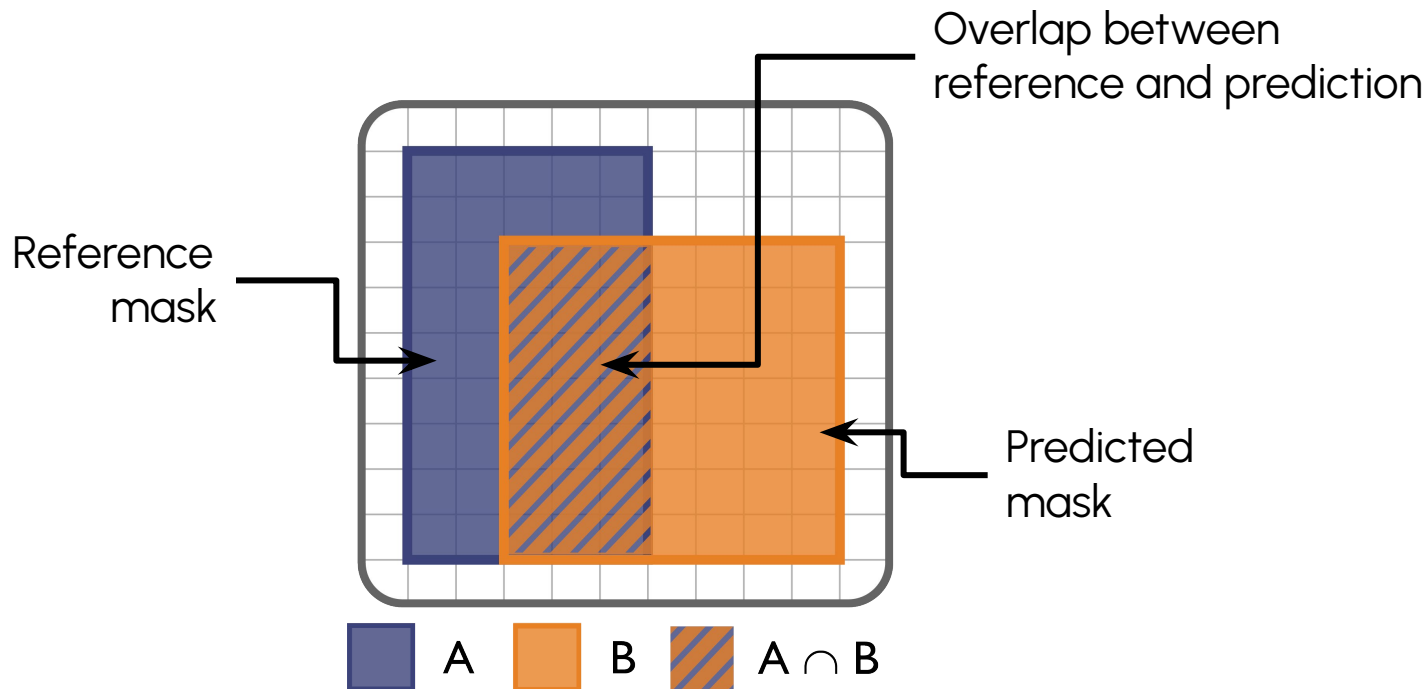


Segmentation

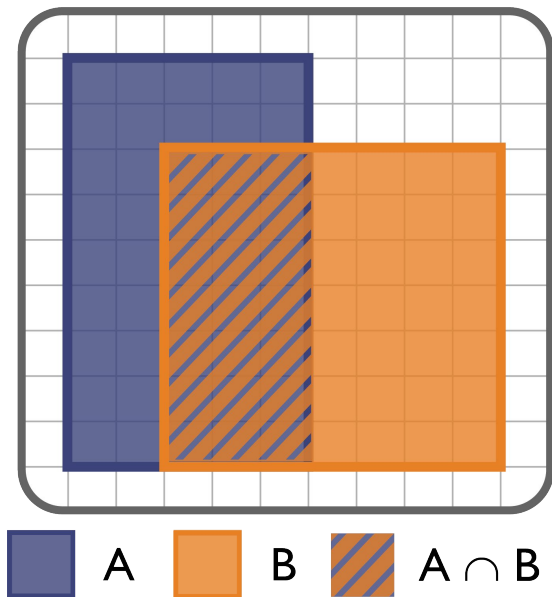
Assign class label(s) per pixel/voxel



Comparison of segmentation masks



Dice Similarity Coefficient (DSC)



$$\text{DSC}(A, B) = \frac{2|A \cap B|}{|A| + |B|}$$

$$\text{DSC}(A, B) = \frac{2 \text{ (Intersection Area)}}{\text{Area of A} + \text{Area of B}}$$

Dice Similarity Coefficient (DSC)



and its relation to other metrics

$$\text{DSC}(A, B) = \frac{2|A \cap B|}{|A| + |B|} \quad \text{IoU}(A, B) = \frac{|A \cap B|}{|A| + |B| - |A \cap B|} = \frac{|A \cap B|}{|A \cup B|}$$

$$\Rightarrow \text{DSC} = \frac{2 \cdot \text{IoU}}{1 + \text{IoU}}$$

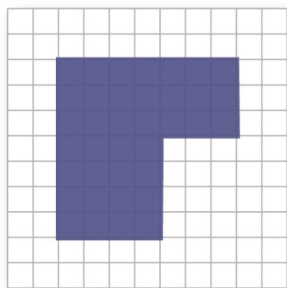
DSC = F1 score

Practical exercise 2

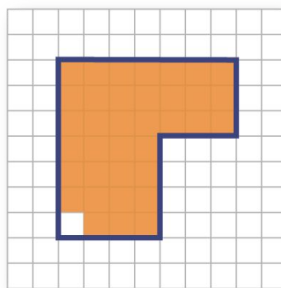


Practical 2

Reference

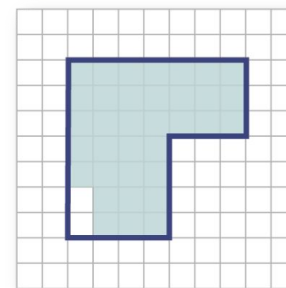


Prediction 1

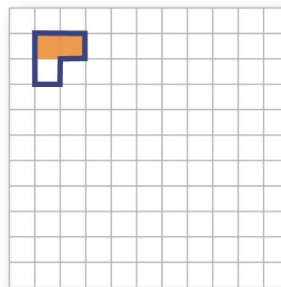
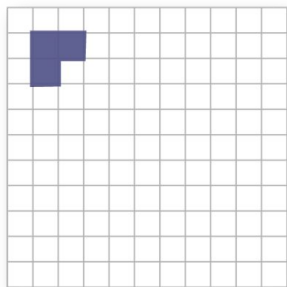


DSC = 0.99

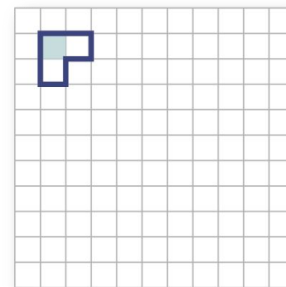
Prediction 2



DSC = 0.97



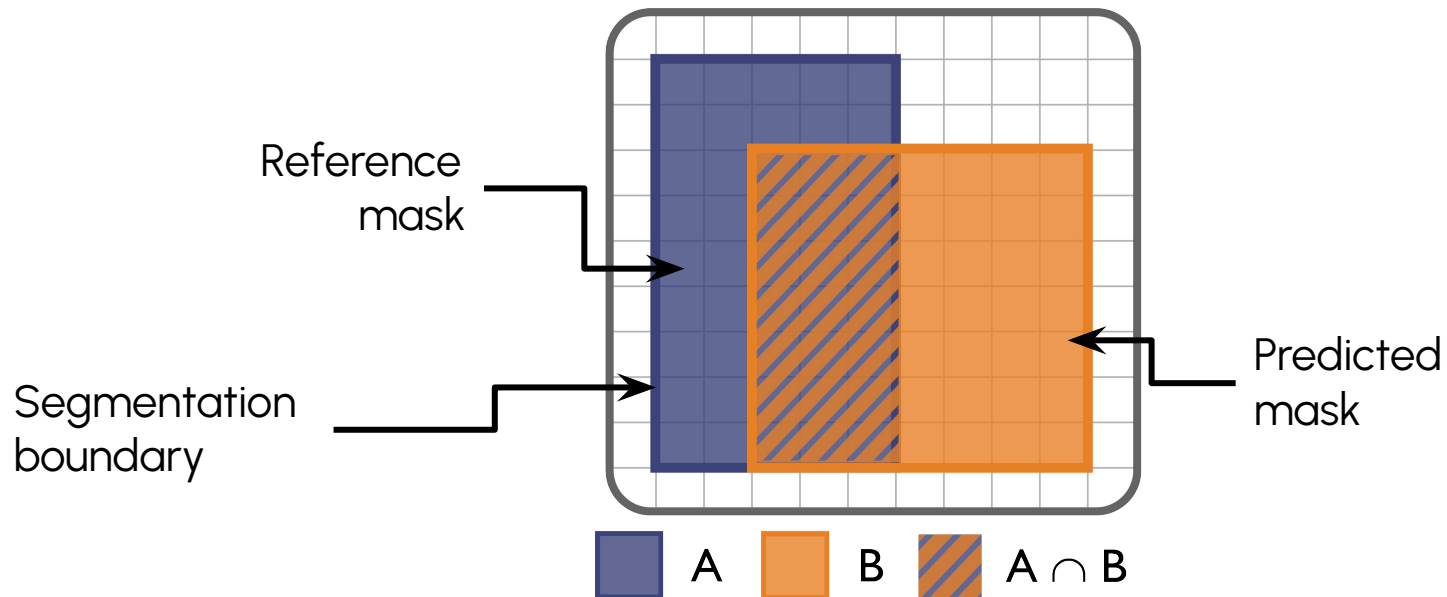
DSC = 0.80



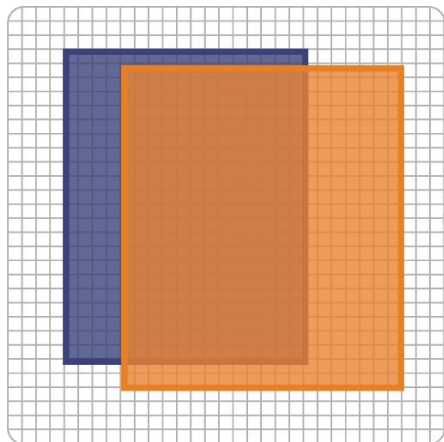
DSC = 0.50



Boundary metrics



Normalized Surface Distance (NSD)

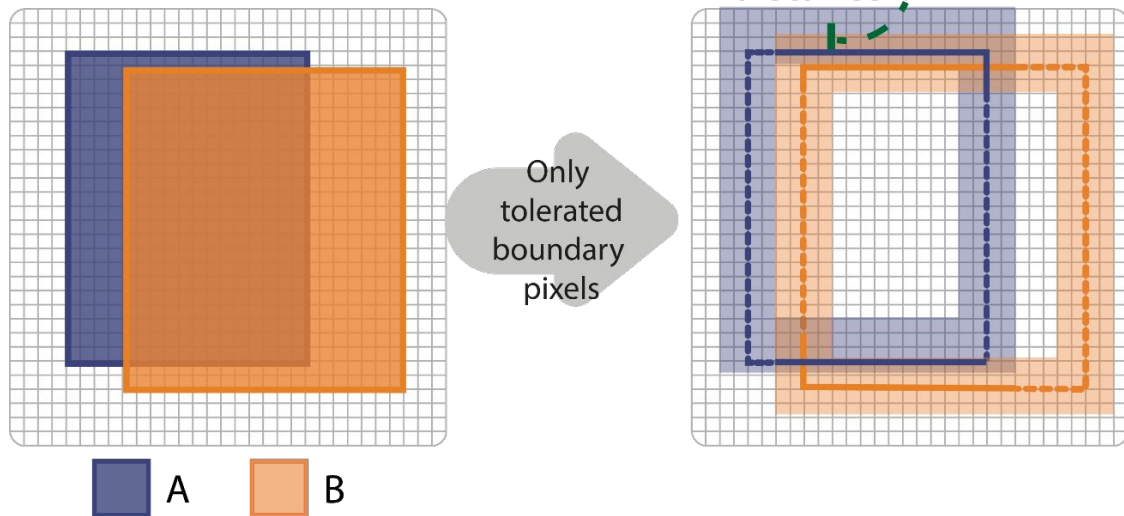


A

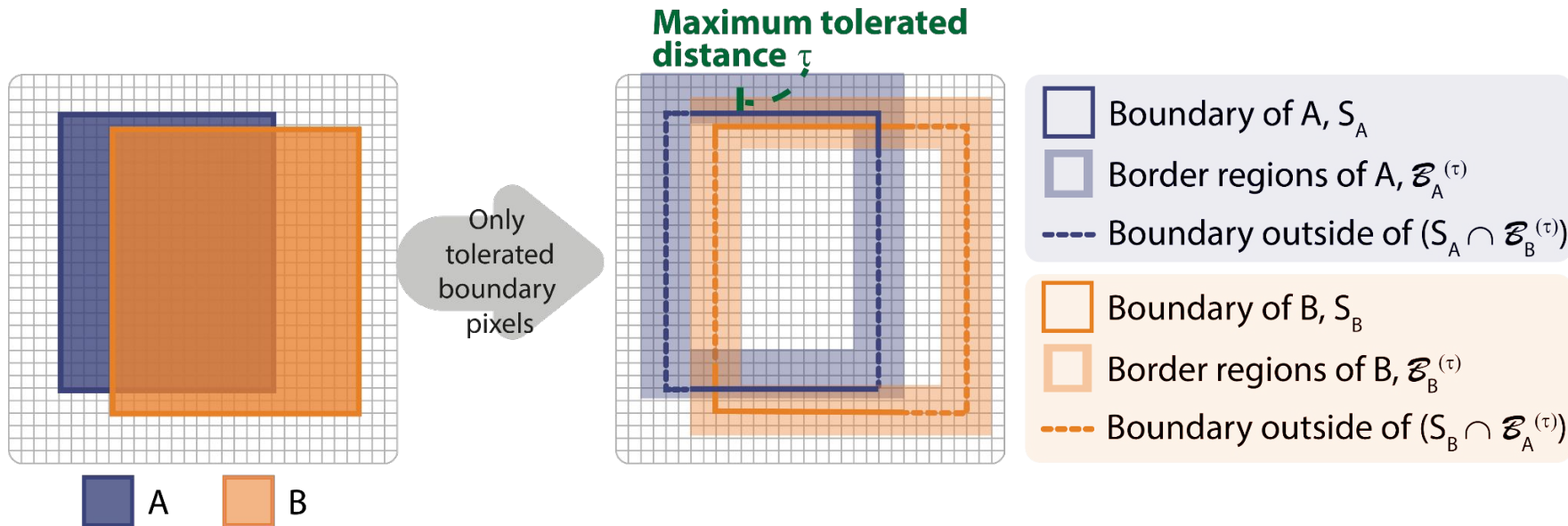


B

Normalized Surface Distance (NSD)

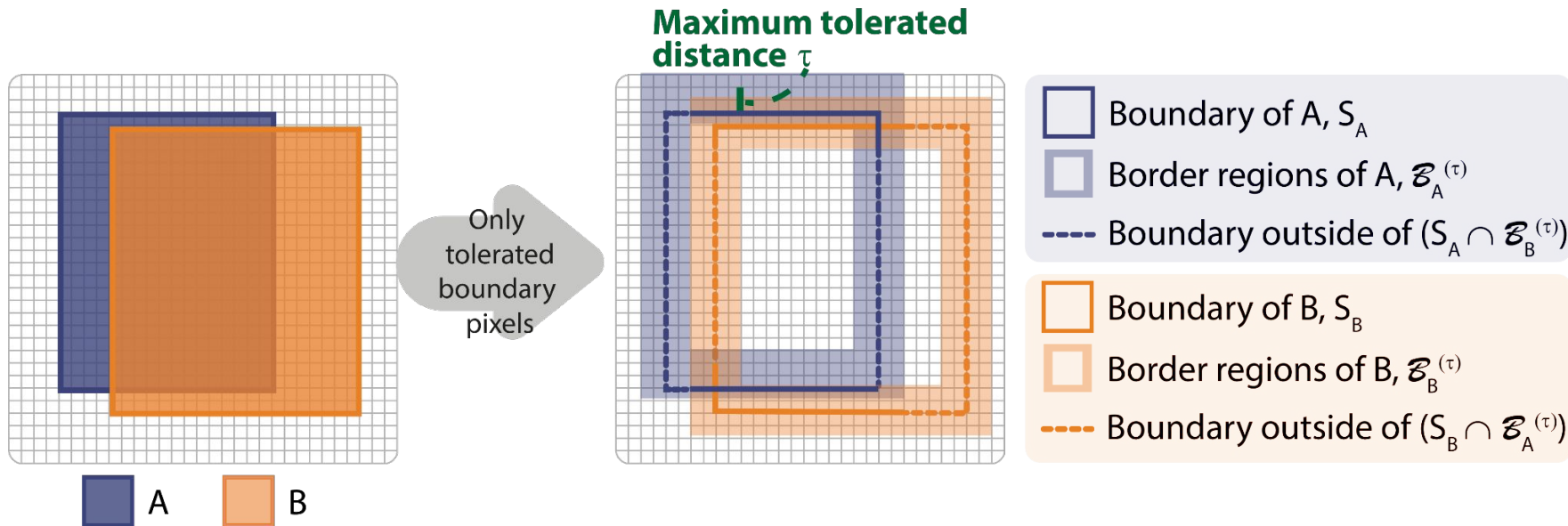


Normalized Surface Distance (NSD)



NSD: DSC on boundary pixels with an uncertainty margin

Normalized Surface Distance (NSD)



$$\text{NSD}(A, B)^{(\tau)} = \frac{\text{[Boundary of A]} + \text{[Boundary of B]}}{\text{[Boundary of A]} + \text{[Boundary of B]}} = \frac{|S_A \cap \mathcal{B}_B^{(\tau)}| + |S_B \cap \mathcal{B}_A^{(\tau)}|}{|S_A| + |S_B|}$$



Metrics Reloaded:

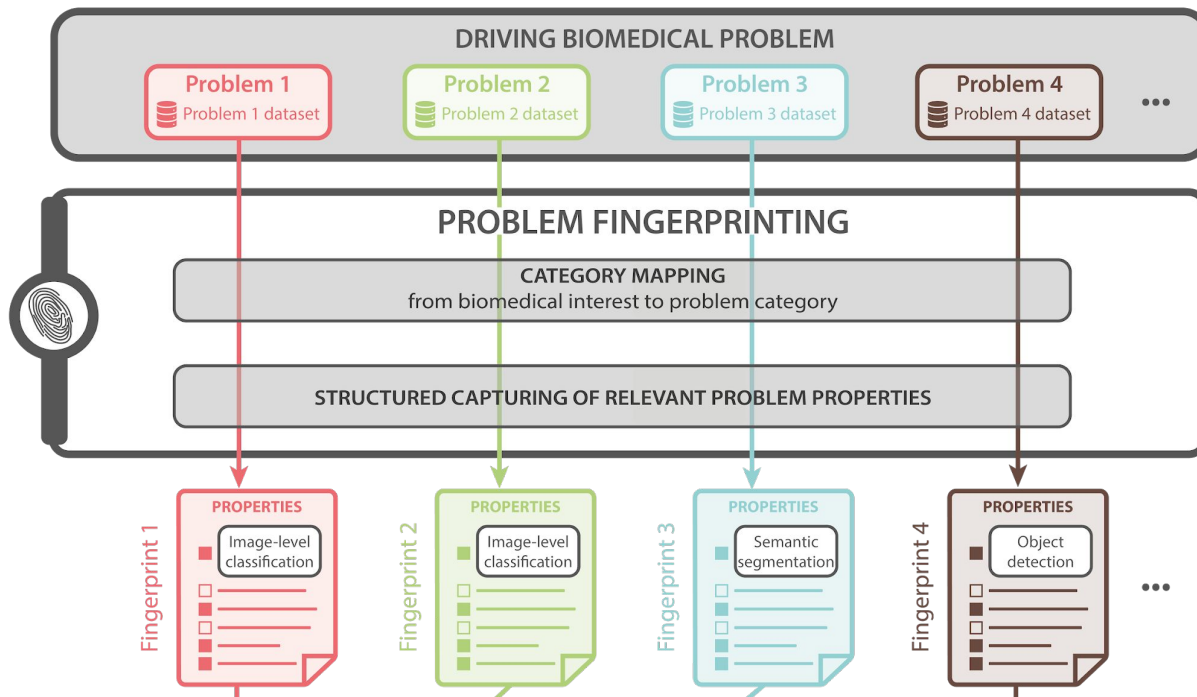
A problem-aware metric
recommendation framework



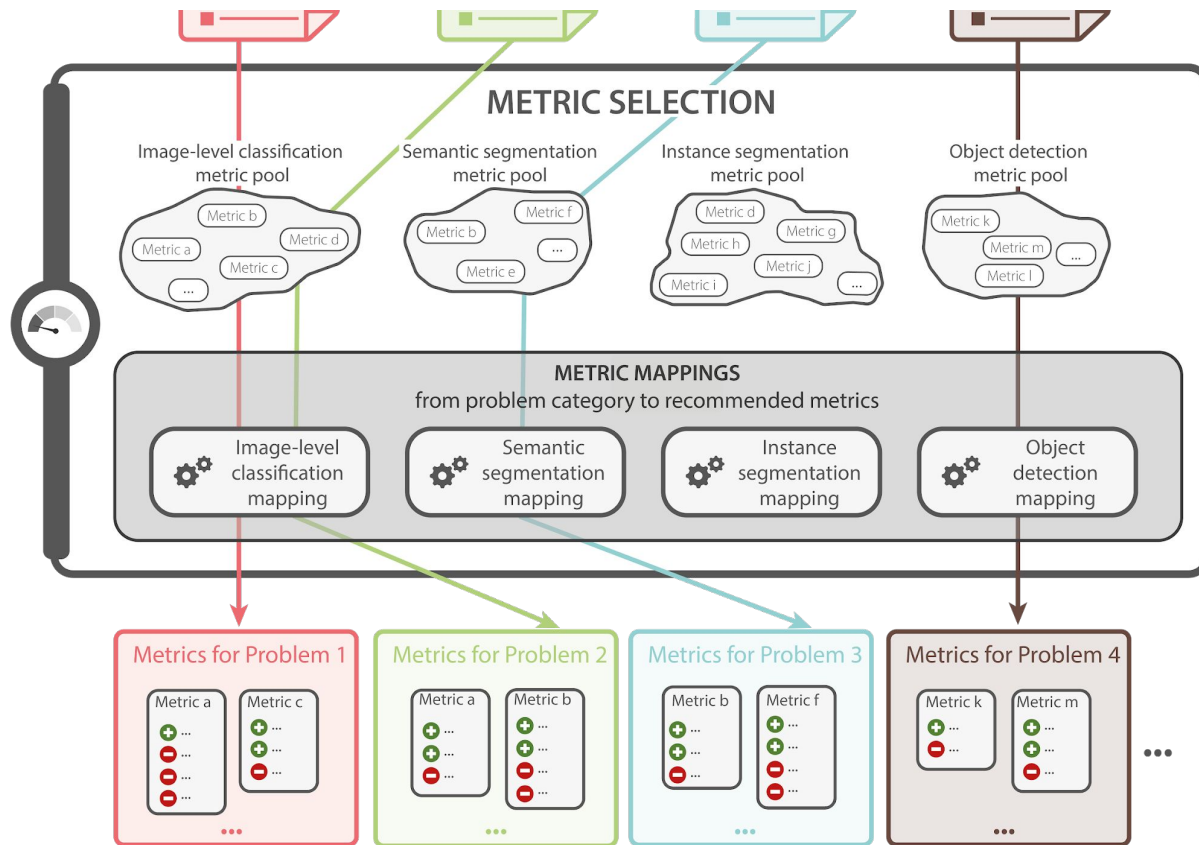
H



Problem-aware metric recommendation framework



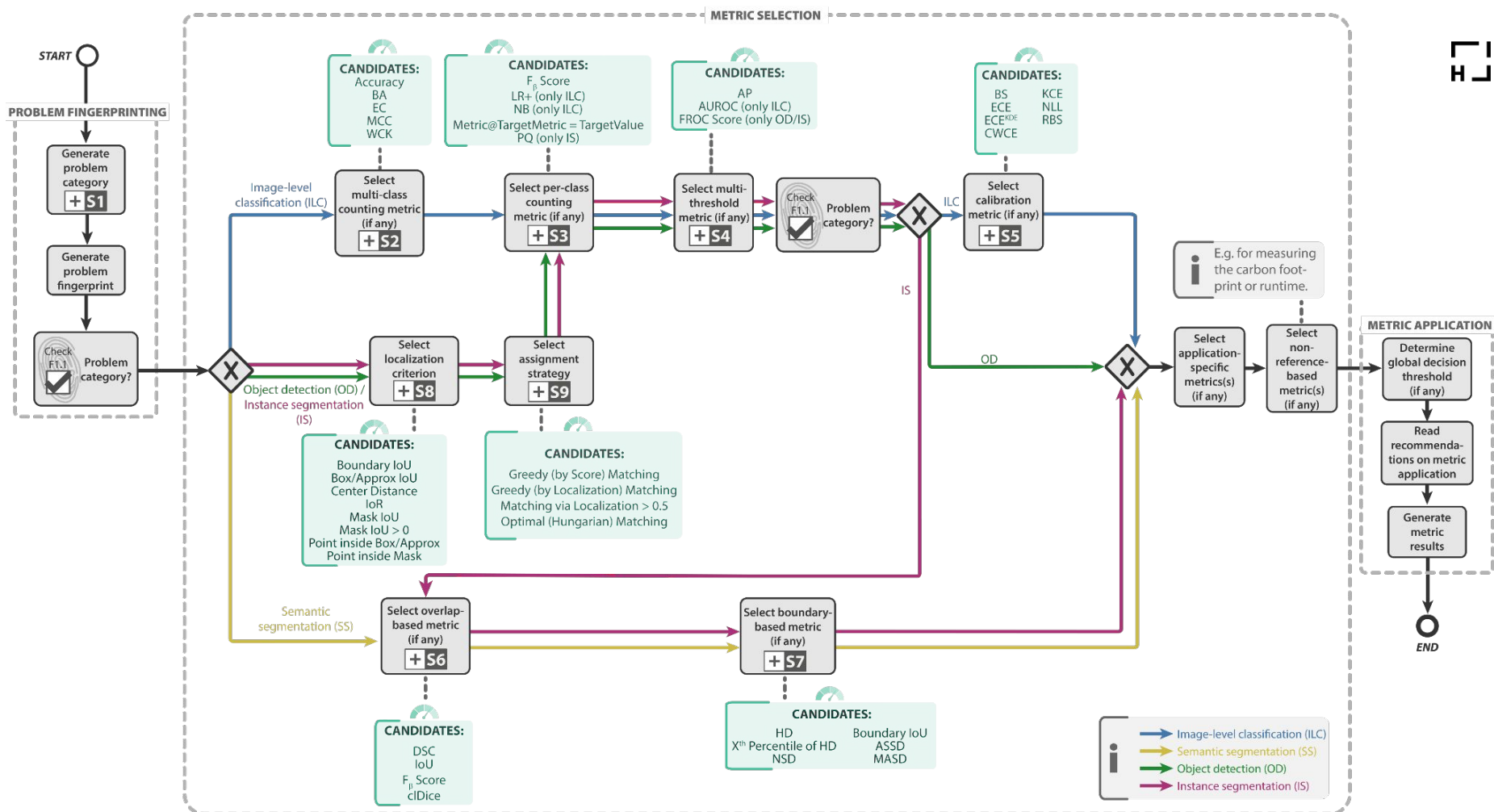
Problem-aware metric recommendation framework

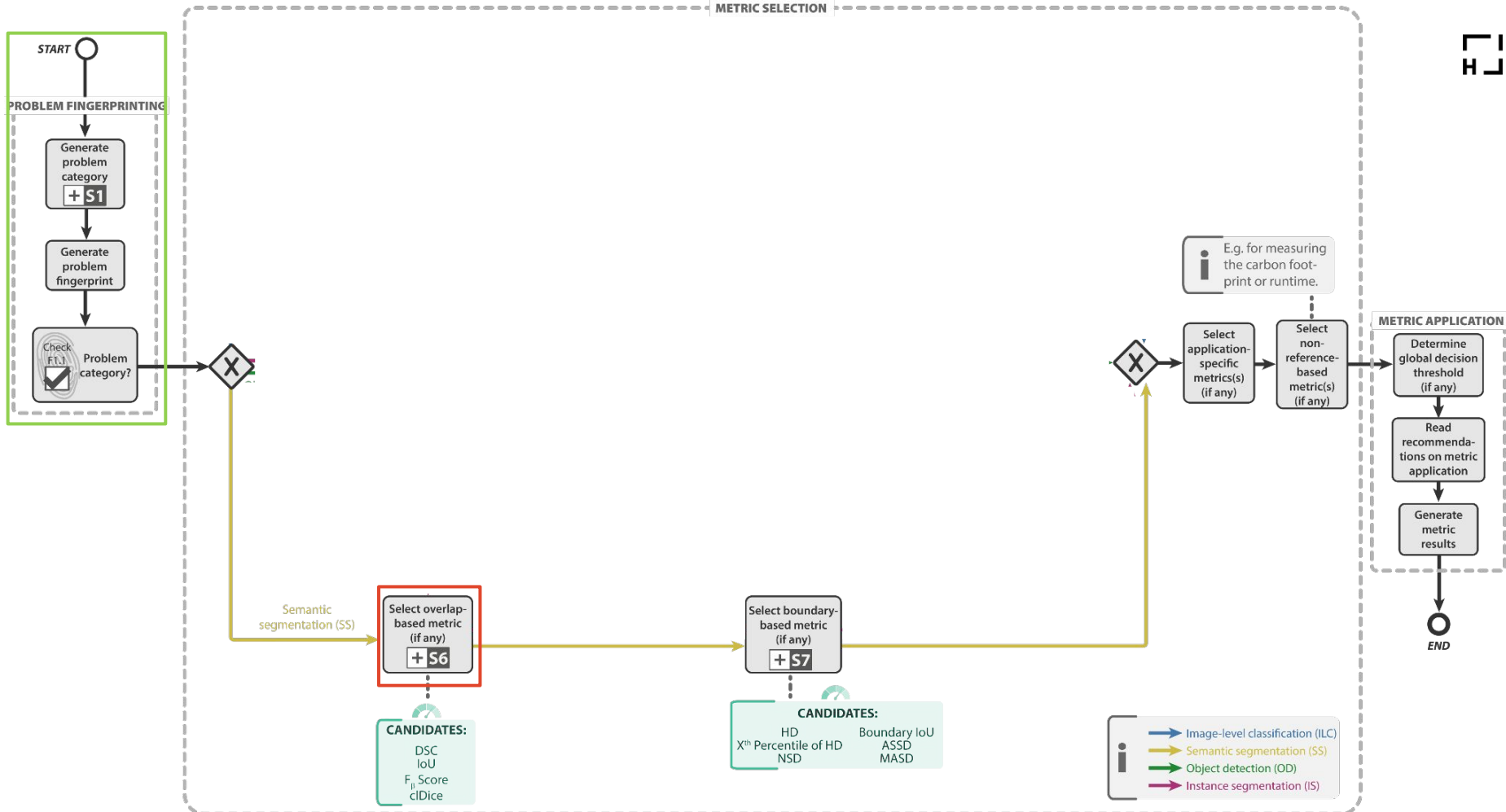


Problem fingerprint



Image processing category identified by category mapping			Semantic segmentation (SS): assignment of one or multiple category labels to each pixel.
Domain interest-related properties (selection)			
Particular importance of structure boundaries			The application requires exact structure boundaries.
Particular importance of structure center (e.g. in cells, vessels)			The application requires accurate knowledge of structure centers.
Compensation for annotation imprecisions requested			The reference annotation is typically only an approximation of the (forever unknown) ground truth. It may be desirable to compensate for known uncertainties such as intra-rater or inter-rater variability, by configuring the metric accordingly. This is only possible for some metrics.
...	...	...	...
Target structure-related properties (selection)			
Small size of structures relative to pixel size			Structures of the provided class are consistently small relative to the grid size in such a way that a single pixel makes up at least several percentage points of the structure volume.
High variability of structure sizes (within one image, across images)			The target structures vary substantially in size, such that some structures are several times the sizes of others.
...	...	...	...
Data set-related properties (selection)			
Presence of class imbalance			The class prevalences differ substantially.
Non-independence of test cases			The test cases are hierarchically structured, indicating non-independence of test cases.
...	...	...	...

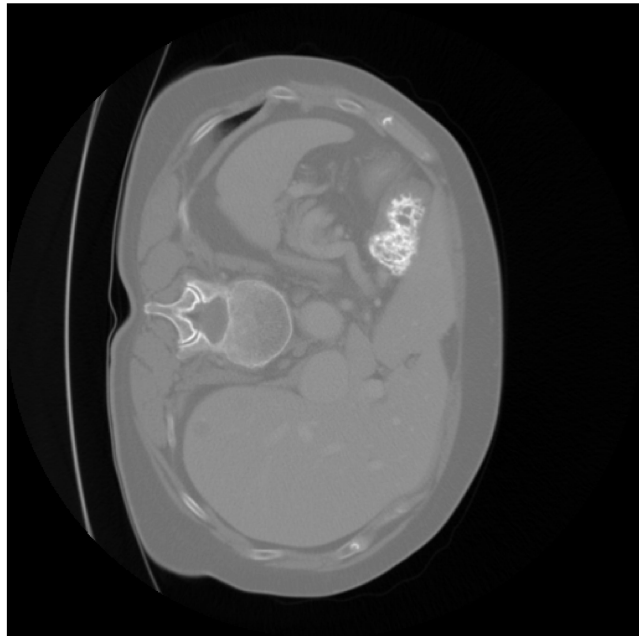




Use case: Liver segmentation



Input



Requested output



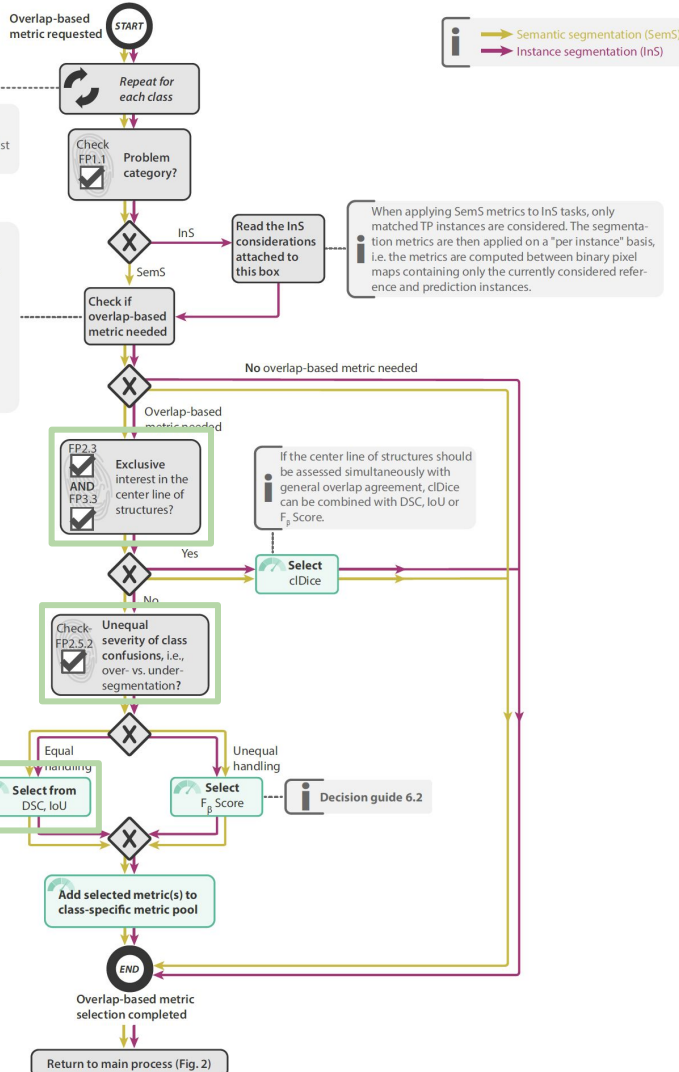
Overlap-based metrics

+ S6

If problem fingerprints differ between classes (e.g., simultaneous segmentation of convex and tubular structures), a class-specific metric pool must be generated (background class: optional).

Overlap metrics are the most popular metric category for segmentation and always recommended for segmentation problems unless the following **EXCLUSION CRITERION** holds:

EXCLUSION CRITERION
 FP3.1 AND FP4.3.1
 Consistently small structures AND noisy reference?



Maier-Hein/Reinke et al. Metrics reloaded:
 Recommendations for image analysis validation.
 Nature Methods 2024

DG6.1: Dice Similarity Coefficient (DSC) versus Intersection over Union (IoU)

Summary of DG6.1: DSC versus IoU

DSC

- ➔ Identical to F_1 Score
- ➔ Close relation to IoU (see Eq. 5)
- ➔ Preference in medical community

IoU

- ➔ Identical to Jaccard Index
- ➔ Close relation to DSC (see Eq. 4)
- ➔ Preference in computer vision community

Extended Data Tab. SN 2.18. Comparison of Dice Similarity Coefficient (DSC) and Intersection over Union (IoU) in the context of the decision guide DG6.1 for Subprocess S6. Context: no exclusive interest in the center line of structures (FP2.3 = FALSE, FP3.3 = FALSE) and equal severity of class confusions (FP2.5.2 = FALSE).

The DSC (Fig. SN 3.5) is identical to the F_1 Score on pixel level and closely related to the IoU (Fig. SN 3.9), which, in turn, is identical to the Jaccard Index (see equations 4 and 5). The two metrics will yield the same ranking of aggregated metric values in most applications (theoretically, deviations are possible), such that there is no value in combining them. Commonly, the computer vision community prefers the IoU, while the medical image community favors the DSC.

$$\text{IoU} = \frac{\text{DSC}}{2 - \text{DSC}} \quad (4)$$

$$\text{DSC} = \frac{2\text{IoU}}{1 + \text{IoU}} \quad (5)$$

Metric cheat sheets

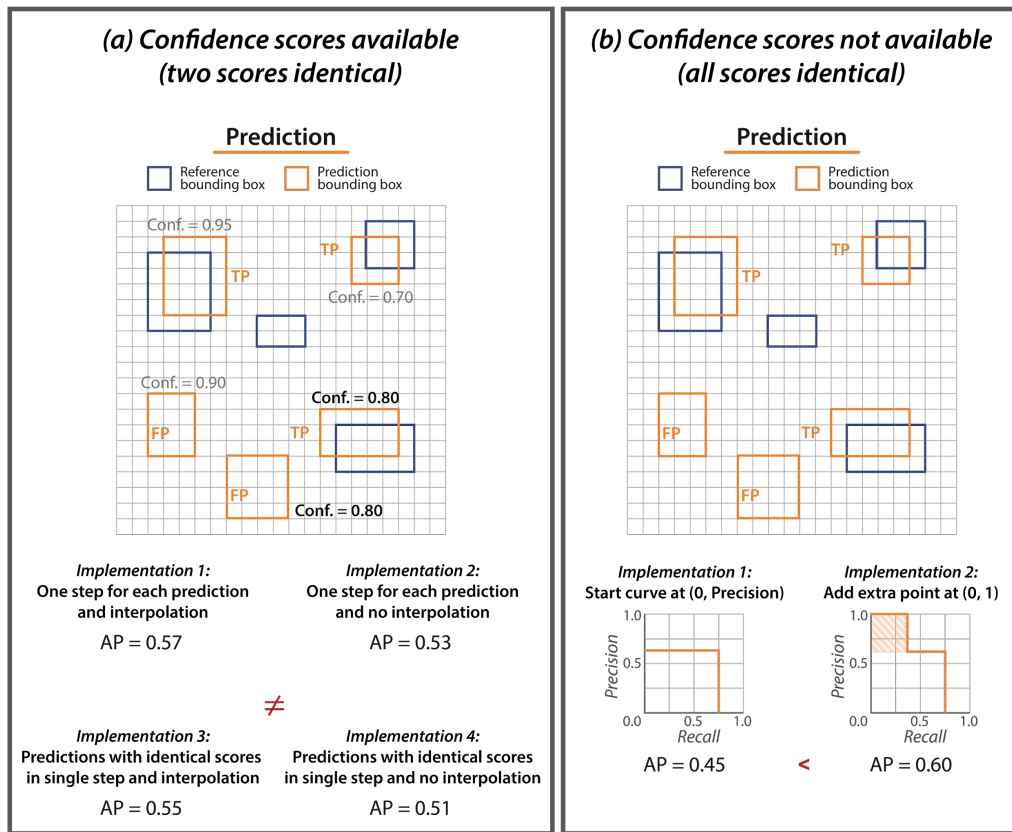


ACCURACY				
$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} = \frac{\text{Correct}}{\text{Total}}$				
VALUE RANGE: [0, 1] ↑				
DESCRIPTION Accuracy measures the ratio of samples that were correctly classified over all predictions made.		DEFINITION [Tharwat, 2020]		
IMPORTANT RELATIONS Accuracy can be rewritten as		METRIC FAMILY Counting metric ☒ Multi-threshold metric ☐ Distance-based metric ☐		
- In binary situations: Accuracy = [Sensitivity • Prevalence + Specificity • (1-Prevalence)] - BA = Accuracy, if classes are balanced - EC = 1 - Accuracy for EC instantiated with 0-1-costs - ER = 1 - Accuracy - (W)CK = 2 • Accuracy - 1, if classes are balanced (and using 0-1-costs) - BM = 2 • Accuracy - 1, if classes are balanced		MULTI-CLASS DEFINITION For C classes, Accuracy is defined as: $\text{Accuracy} = \frac{1}{N} \sum_{i=1}^C n_{ii}$ n_{ii}: diagonal entries of the confusion matrix; sum equals number of correctly classified samples N: total number of samples		
PREVALENCE DEPENDENCY ☒		RECOMMENDED FOR ImLC ☐ SemS ☐ ObD ☐ InS ☐		
CARDINALITIES TP ☒ FP ☒ FN ☒ TN ☒				
RELEVANT PITFALLS - Accuracy is highly sensitive to class imbalance (Figs. 7a, 24 in [Reinke et al., 2023]). - Accuracy is prevalence-dependent, thus not comparable across data sets with different prevalences (Fig. 16 in [Reinke et al., 2023]). - Accuracy does not allow for incorporating unequal severity of confusions across classes. This implies that the metric is not applicable to cost-benefit analyses (Fig. 18 in [Reinke et al., 2023]), or when target classes are related on an ordinal scale (Fig. 5b in [Reinke et al., 2023]). - Accuracy depends on the definition of TN (undefined for ObD and InS).				
RECOMMENDATIONS - Accuracy should generally not be considered... • ... in the presence of class imbalance unless class prevalences reflect the interest across classes. • ... if comparison of performance across data sets with different prevalences is desired. • ... if class confusions are of unequal severity (examples: ordinal target classes, cost-benefit analysis). - Due to ease of interpretation and popularity, we specifically recommend it as a multi-class metric if the compensation of class imbalances is not desired.				

BALANCED ACCURACY (BA)	
$BA = \frac{1}{2} (\text{Sensitivity} + \text{Specificity}) = \frac{1}{2} \left(\frac{\text{TP}}{\text{TP} + \text{FN}} + \frac{\text{TN}}{\text{TN} + \text{FP}} \right)$	
VALUE RANGE: [0, 1] ↑	
DESCRIPTION BA measures the arithmetic mean of Sensitivities for each class, i.e., for each class, it measures the fraction of actual positive samples that were predicted as such.	METRIC FAMILY Counting metric ☒ Multi-threshold metric ☐ Distance-based metric ☐
DEFINITION [Tharwat, 2020]	PREVALENCE DEPENDENCY ☐
IMPORTANT RELATIONS - J = 2BA - 1 - (W)CK = 2BA - 1, if classes are balanced (and using 0-1-costs) - Accuracy = BA, if classes are balanced - EC = 1 - BA, if EC costs are chosen such that $w_{ij} = 0$ and $w_{ji} = 1/C_{ij}$, where w_{ij} are the costs for a sample of actual class i that was predicted as class j, C is number of classes and P_i is prevalence of class i.	MULTI-CLASS DEFINITION For C classes, BA is defined as the arithmetic mean of Sensitivities per class: $BA = \frac{1}{C} \sum_{i=1}^C \text{Sensitivity}_i = \frac{1}{C} \sum_{i=1}^C \frac{n_{ii}}{n_{i+}}$ n_{i+}: diagonal entries of the confusion matrix; sum equals number of correctly classified samples n_{i+}: sum of entries of row i in the confusion matrix N: total number of samples
CARDINALITIES TP ☒ FP ☒ FN ☒ TN ☒	RECOMMENDED FOR ImLC ☐ SemS ☐ ObD ☐ InS ☐
RELEVANT PITFALLS - BA can be misleading for imbalanced situations (Fig. 7a in [Reinke et al., 2023]). - BA does not allow for incorporating unequal severity of confusions across classes. This implies that the metric is not applicable to cost-benefit analyses (Fig. 18) in [Reinke et al., 2023], or when target classes are related on an ordinal scale (Fig. 5b in [Reinke et al., 2023]). - BA is not well-suited if an unequal treatment of classes is requested (e.g., some classes are treated as more important than others) [Grandini et al., 2020]. - BA is insensitive to changes in predictive values (PPV and NPV) [Maier-Hein et al., 2022]. - In binary tasks, BA may yield the same value for different Sensitivity and Specificity scores [Reinke et al., 2021]. - BA depends on the definition of TN (undefined for ObD and InS).	
RECOMMENDATIONS - BA should not be applied if... • ... there is an unequal interest across classes. • ... predictive values should be assessed. • ... class confusions are of unequal severity (examples: ordinal target classes, cost-benefit analysis). - Otherwise, it should specifically be considered... • ... in the presence of high class imbalance in case there is an equal interest across classes. • ... if a comparison of performance across data sets with different prevalences is desired. - BA can be used to identify and validate the decision rule applied to predicted class scores.	

MATTHEWS CORRELATION COEFFICIENT (MCC)					
Synonyms: Phi Coefficient					
$\text{MCC} = \frac{\text{TP} \cdot \text{TN} - \text{FP} \cdot \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}} = \frac{\text{Correct} - \text{Wrong}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}$					
VALUE RANGE: [-1, 1] ↑ A value of 0 refers to a prediction which is not better than random guessing.					
DESCRIPTION MCC measures the correlation between the actual and the predicted class.			DEFINITION [Matthews, 1975]		
IMPORTANT RELATIONS MCC can be rewritten as: $\text{MCC} = \sqrt{\text{PPV} \cdot \text{Sensitivity} \cdot \text{Specificity} \cdot \text{NPV}} = \sqrt{(1 - \text{PPV}) \cdot (1 - \text{Sensitivity}) \cdot (1 - \text{Specificity}) \cdot (1 - \text{NPV})}$ MCC is equivalent to the geometric mean of Markedness and Informedness.			RECOMMENDED FOR ImLC ☐ SemS ☐ ObD ☐ InS ☐		
MULTI-CLASS DEFINITION For C classes, MCC can be defined as: $\text{MCC} = \frac{\sum_{i=1}^C (\sum_{j=1}^C n_{ij} - n_{ii}) \cdot \sum_{k=1}^C (\sum_{l=1}^C n_{kl} - n_{kk})}{\sqrt{(\sum_{i=1}^C (\sum_{j=1}^C n_{ij} - n_{ii})^2) (\sum_{k=1}^C (\sum_{l=1}^C n_{kl} - n_{kk})^2)}}$ n_{ij}: entry of the confusion matrix for row i and column j, i.e., samples of actual class i that were predicted as class j			PREVALENCE DEPENDENCY ☒		
RELEVANT PITFALLS - MCC is prevalence-dependent, thus not comparable across data sets with different prevalences (Fig. 16 in [Reinke et al., 2023]). - MCC does not allow for incorporating unequal severity of confusions across classes. This implies that the metric is not applicable to cost-benefit analyses (Fig. 18 in [Reinke et al., 2023]), or when target classes are related on an ordinal scale (Fig. 5b in [Reinke et al., 2023]). - The theoretical lower bound of MCC (-1) may not always be achievable (Fig. 44 in [Reinke et al., 2023]). - MCC is hard to interpret [Zhu 2020]. - Compared to other metrics like EC, MCC lacks a framework to identify and validate the decision rule applied to predicted class scores [Ferrer 2022]. - MCC depends on the definition of TN (undefined for ObD and InS).			METRIC FAMILY Multi-threshold metric ☐ Distance-based metric ☐		
RECOMMENDATIONS - MCC should not be used/used with care if... • ... class confusions are of unequal severity (example: ordinal target classes). • ... the provided class prevalences do not reflect the population of interest. • ... there is a mismatch between class prevalences and class importance. • ... compensation for class imbalance is not requested. - Otherwise, MCC should be used as a multi-class metric specifically if all basic error rates (Sensitivity, Specificity, PPV, NPV) should be captured in one score. - MCC scores should be carefully interpreted in the presence of class imbalance as the distribution becomes skewed [Zhu 2020].					

Pitfall: Different implementations – different results



Standardized metric implementation



Metrics Reloaded

A Python implementation of [Metrics Reloaded](#) - A new recommendation framework for biomedical image analysis validation.

docs passing Unit Tests passing codecov 62%

Installation

Using git

Create and activate a new [Conda](#) environment:

```
conda create -n metrics python=3.10 pip
conda activate metrics
```

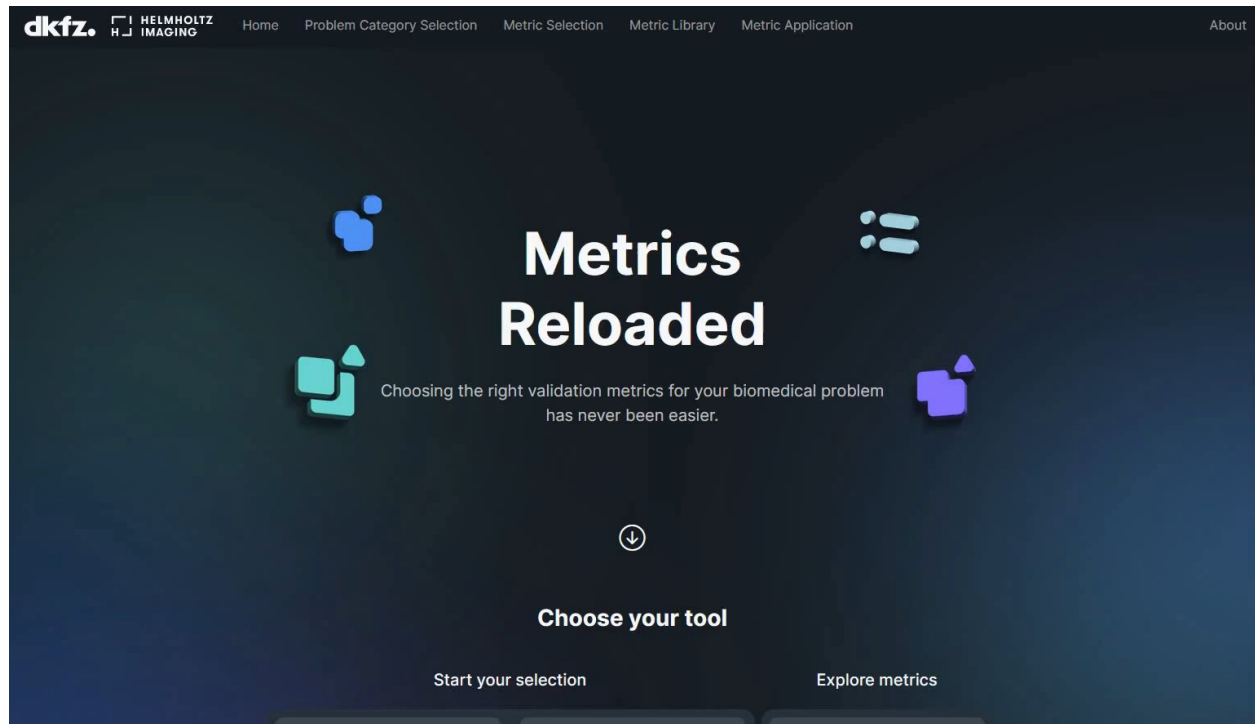
Clone the repository:

```
git clone https://github.com/Project-MONAI/MetricsReloaded.git
```



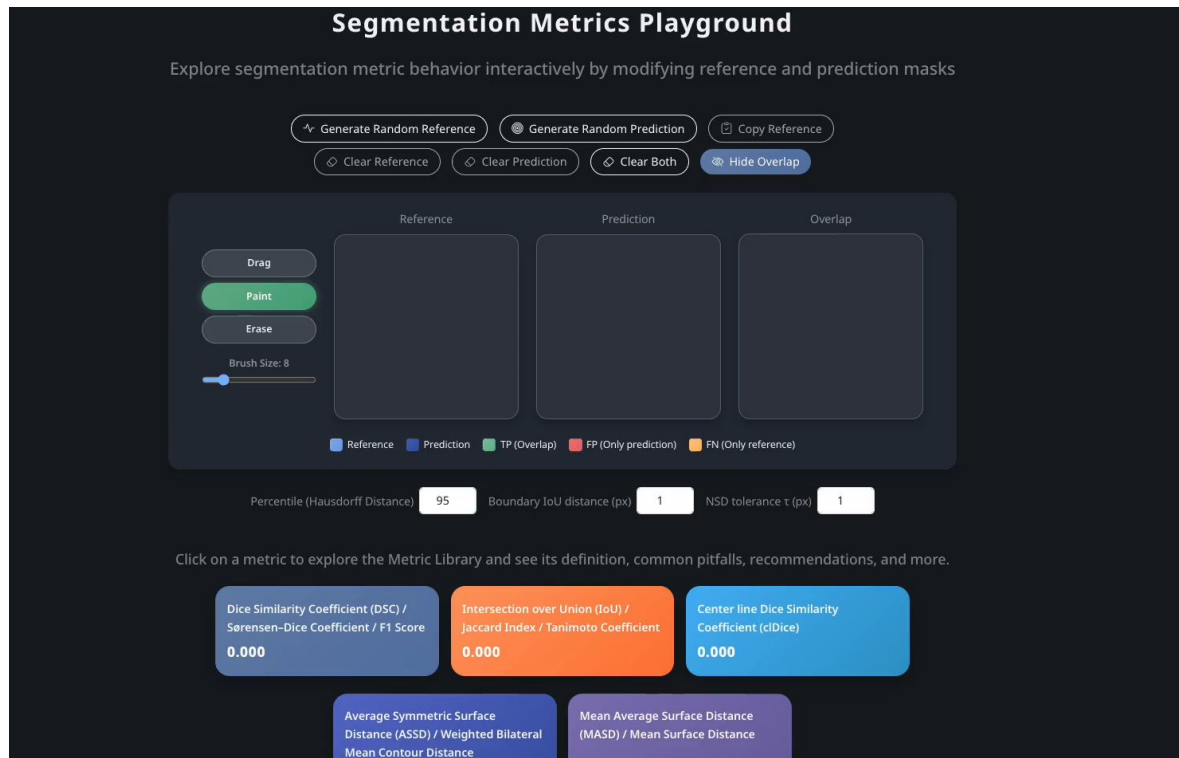
<https://github.com/Project-MONAI/MetricsReloaded/tree/main/MetricsReloaded>

Metrics Reloaded toolkit



<https://metrics-reloaded.dkfz.de>

Metrics Reloaded toolkit



<https://metrics-reloaded.dkfz.de/metric-demo>

Practical exercise 3

Breakout room 1 - Use Case 1

Breakout room 2 - Use Case 2

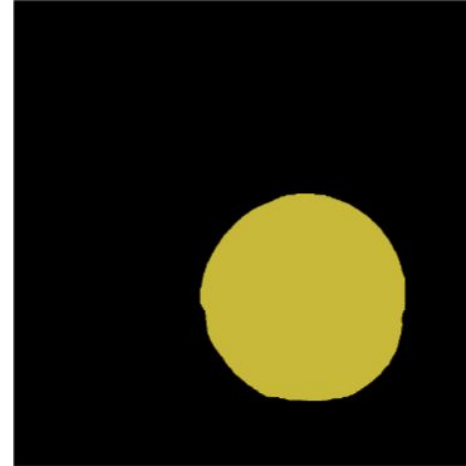
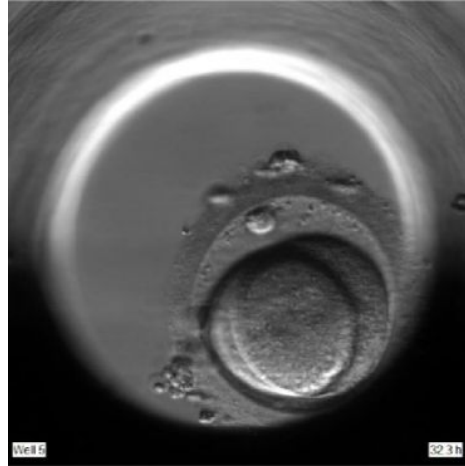
...

Use Case 1: AI-assisted skin lesion diagnosis

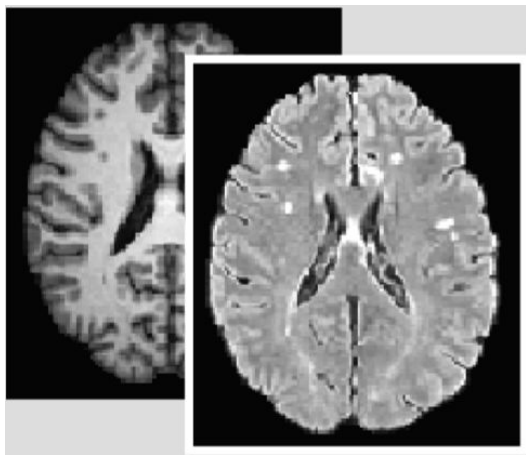


Dermatofibroma: 0.6
Melanocytic nevus: 0.2
Melanoma: 0.1
Basal cell carcinoma: 0.0
Actinic keratosis: 0.0
Benign keratosis: 0.0
Vascular lesion: 0.1

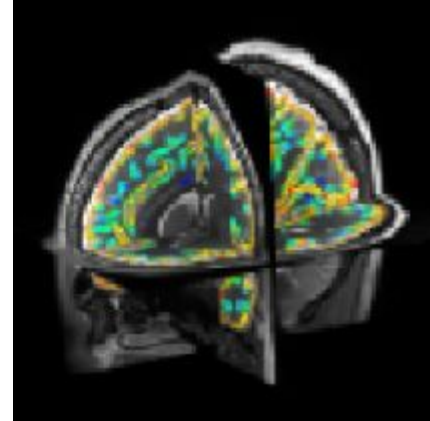
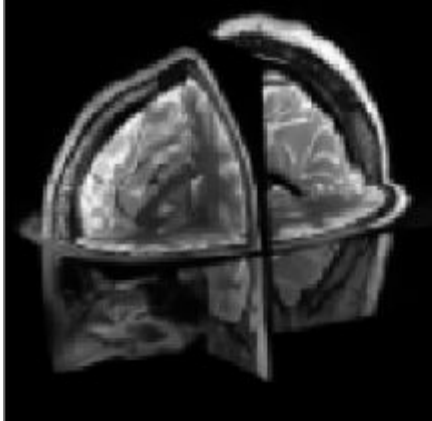
Use Case 2: Segmentation of lung cancer cells in microscopy images



Use Case 3: Identification of new multiple sclerosis lesions in follow-up MRI scans



Use Case 4: Mapping cortical structures in brain MRI

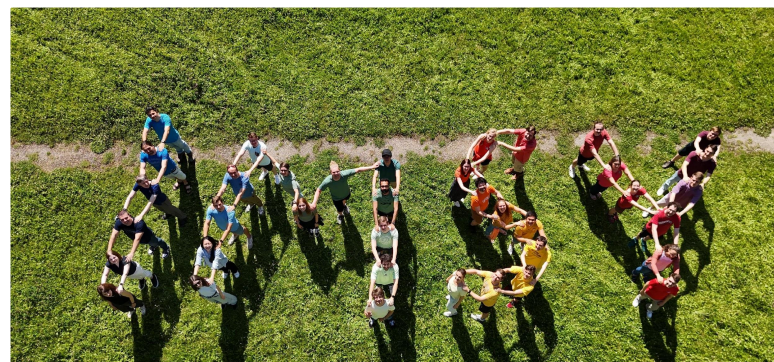


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